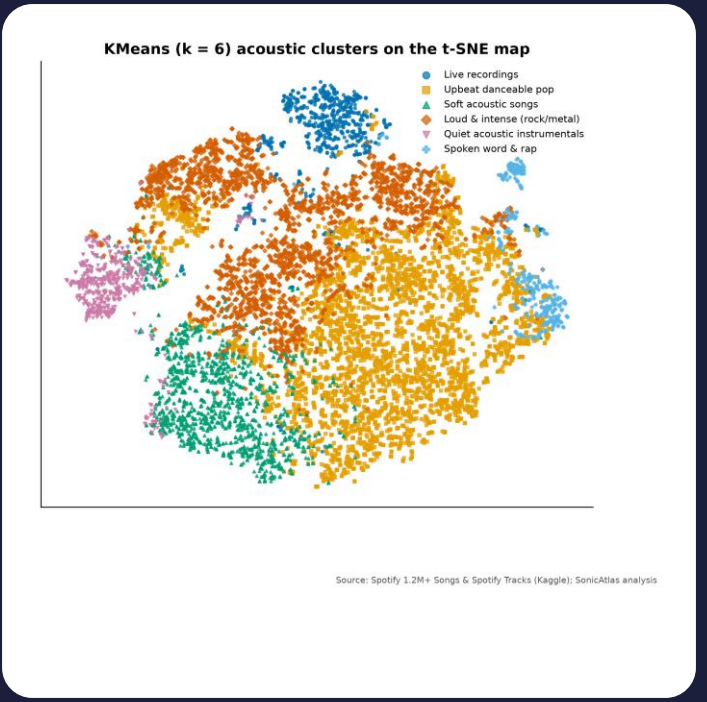
SONICATLAS

An Interactive In-Browser EDA Platform for Spotify Audio Features and Lyrics Across Genres and Eras

Review-1 | Problem, literature and design

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Real acoustic map of 10,000 tracks (t-SNE)

BACSE301 Exploratory Data Analysis (Project-based Course)
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ABSTRACT

Mapping how the sound of music changes across genres and eras

Spotify's archived audio descriptors (energy, valence, danceability, acousticness and more) are rarely explored beyond static charts. SonicAtlas cleans 1,204,025 tracks released 1900-2020, a genre table of 89,740 tracks in 114 genres and 161,401 English lyrics, explores them with the course methods (descriptive statistics, correlation, PCA/t-SNE/LLE, time series, hypothesis tests, text mining), and publishes everything in a website that runs SQL inside the browser with DuckDB-WASM.

1,204,025

tracks analysed

114

genres compared

1959-2020

years of trends

Prior work is static and studies sound and words apart

Study	What they did	Gap SonicAtlas addresses
J. Serrà et al. (2012)	Tracked pitch, timbre and loudness of Western pop over 50+ years	Static figures; no genre-level or interactive view
M. Interiano et al. (2018)	Linked audio features of 500k UK releases to chart success	No lyrics; results cannot be re-sliced by readers
C. O. Brand et al. (2019)	Measured rising negativity in song lyrics over 50 years	Lyrics only; audio mood not compared
L. van der Maaten et al. (2008)	Introduced t-SNE for 2-D maps of high-dimensional data	Global distances unreliable; no music benchmark
D. H. da Silva et al. (2025)	Streamlit web app for EDA and ML on medical data	Needs a server; not music-aware

Static results

fixed figures, no re-slicing

Sound vs words

audio and lyrics never compared

Data quality

zero tempos, re-issues ignored

Generic tools

not music-aware, need a server

25 verified papers reviewed in the report (Table I).

PROBLEM IDENTIFICATION

Open music data exists, but it is messy and locked in static charts

Messy data

2,777 zero-BPM tempos, 71,694 duplicate title+artist rows, no shared keys

Frozen results

papers and notebooks show fixed figures; readers cannot ask new questions

Split modalities

audio features and lyrics are analysed in isolation

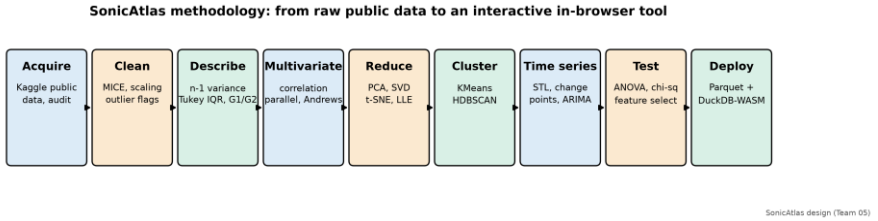
Hosting cost

interactive tools usually need a running server

OBJECTIVES

Seven measurable objectives

1. Clean 1M+ tracks; validate MICE by masking 5% of values
2. Descriptive stats in course conventions (n-1, Tukey IQR, G1, G2)
3. Correlation and multivariate genre fingerprints
4. Compare PCA, t-SNE, LLE, UMAP; name acoustic clusters
5. Trends, STL seasonality, change points, forecast
6. ANOVA, chi-square, feature selection; lyrics sentiment
7. In-browser SQL with lite-data queries under 1 second



Existing systems cover only parts of the need

System	Open data	Music-specific EDA	Audio + lyrics	Interactive	Serverless
Spotify app / Web API	No (closed)	No	No	Yes (app)	No
Static notebooks / papers	Yes	Partly	Rarely	No	No
Streamlit EDA app [25]	Yes	Generic	No	Yes	No (server)
YData / Sweetviz / D-Tale	Yes	Generic	No	Partly	No (server for D-Tale)

No existing option is open, music-aware, interactive and serverless at the same time.

PROPOSED SYSTEM

A documented pipeline plus a serverless query platform

Clean

MICE for impossible values, re-issue and outlier flags, scaling

Explore

statistics, correlation, parallel coordinates, Andrews curves

Map

PCA, SVD, t-SNE, LLE, UMAP and named clusters

Explain

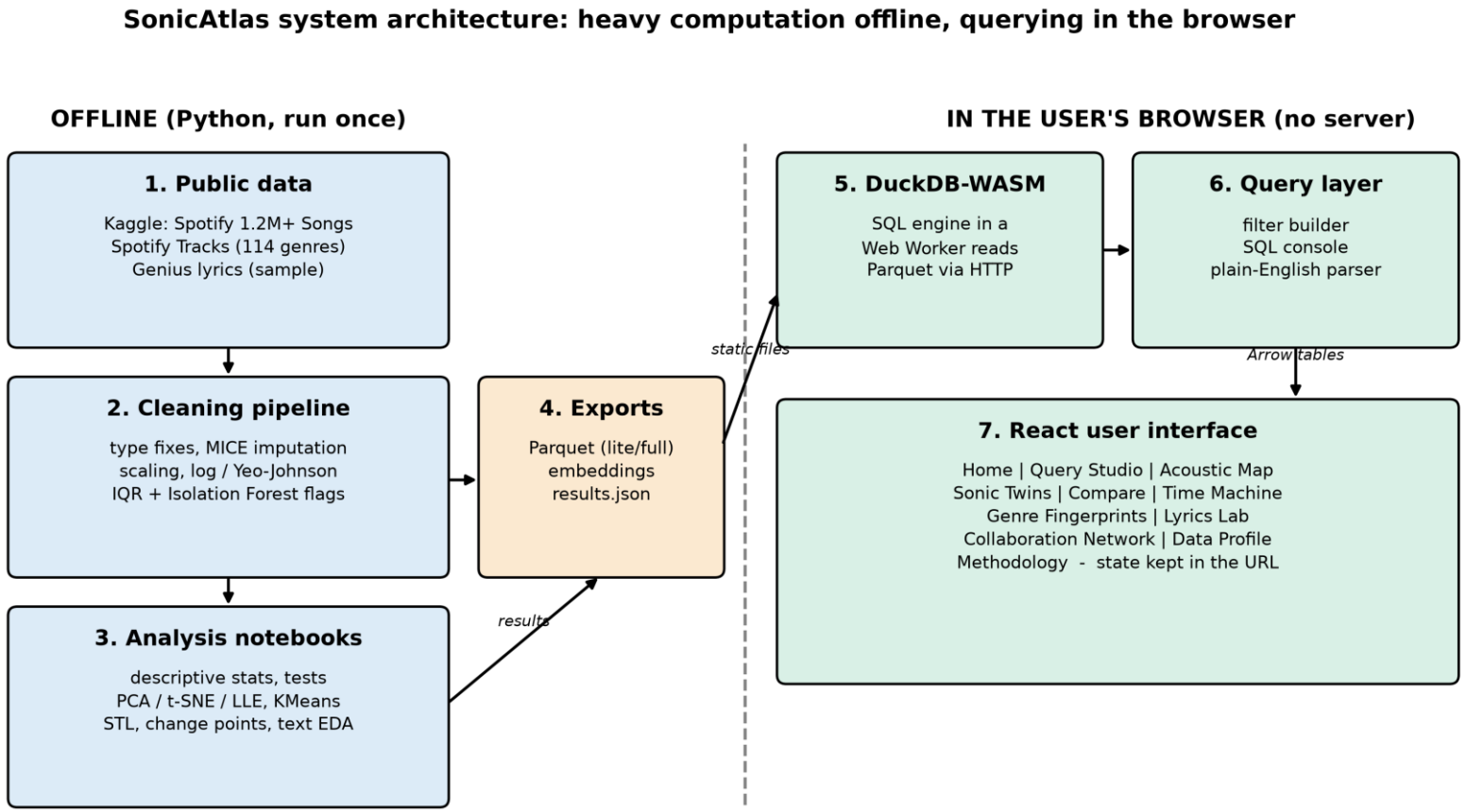
STL, change points, ANOVA, chi-square, lyrics sentiment

Share

static website: DuckDB-WASM SQL, maps, comparisons, URLs

Heavy computation runs once in Python; every interaction runs locally in the user's browser.

Heavy maths runs offline; querying runs in the user's browser



SonicAtlas design (Team 05)